Decision Making with Sentiment Analysis in R: From Social Media Noise to Managerial Action

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Abstract

This handout explains how sentiment analysis can support managerial decision-making. It introduces the basic idea of sentiment analysis, presents a reproducible R workflow, and demonstrates how airline-related tweets can be transformed into decision-relevant insights. The example uses tidyverse and tidytext to clean, tokenize, visualize, and interpret customer feedback.

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0.1 Online Access

Access the Handout

- **Handout:** https://masimruhi.github.io/dadm/index.html
- **Presentation:** https://masimruhi.github.io/dadm/presentation.html
- **Repository:** https://github.com/masimruhi/dadm

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1 Introduction

Modern companies receive large amounts of customer feedback through social media, online reviews, support chats, and survey comments. Much of this feedback is unstructured text, which means that it cannot be analysed as directly as numerical variables such as sales, prices, or ratings. Sentiment analysis helps convert this kind of text into structured information that can support decision-making (Liu, 2012; Pang & Lee, 2008).

This handout focuses on a practical question: How can customer tweets be transformed into decision-relevant insights for airline management? The example uses airline-related tweets and demonstrates a reproducible workflow in R. The goal is not only to classify text as positive or negative, but to connect the result to managerial action.

2 What is Sentiment Analysis?

Sentiment analysis is a text analysis method that identifies emotional orientation in written language. In a simple form, the method classifies text as positive, negative, or neutral. More advanced applications can detect emotions, topics, intensity, or changes over time. In business contexts, sentiment analysis is commonly used to monitor customer satisfaction, brand reputation, product feedback, and service quality.

A typical workflow includes importing text data, cleaning it, splitting the text into smaller units such as words, removing irrelevant words, matching words with sentiment dictionaries, and summarising the results. In R, packages such as `tidyverse` and `tidytext` make this workflow relatively accessible and reproducible (Silge & Robinson, 2017; Wickham et al., 2023).

3 Why is it Relevant for Decision-Making?

Decision-makers often need to identify which problems deserve attention first. When feedback is scattered across thousands of tweets, manual reading becomes slow and inconsistent. Sentiment analysis can quickly summarize the emotional tone of customer comments and highlight repeated sources of dissatisfaction.

For example, if an airline receives many negative comments about delayed flights, management can improve delay communication, operational punctuality, or compensation processes. If most negative comments refer to customer service, management may prioritize staff training, support capacity, or service recovery. The value of sentiment analysis therefore depends on whether the results lead to concrete decisions.

There are different approaches to sentiment analysis, including lexicon-based methods, machine learning methods, transformer-based models, and hybrid approaches. This project focuses on a lexicon-based approach because it is transparent, reproducible in RStudio, and suitable for a short live demonstration.

4 Data and Example Context

The example uses the Twitter US Airline Sentiment dataset, which contains 14,640 airline-related tweets. The dataset includes the tweet text, the airline, a sentiment label, and, for negative tweets, a reason for the negative sentiment.

The following code summarizes the distribution of the provided sentiment labels.

```r
tweets |>
  count(airline_sentiment) |>
  mutate(share = scales::percent(n / sum(n))) |>
  knitr::kable()
```

airline_sentiment	n	share
negative	9178	62.7%
neutral	3099	21.2%
positive	2363	16.1%

This structure makes the dataset useful for a classroom example. It contains both unstructured text and already labelled sentiment categories. The labels make it possible to compare simple visual summaries with a dictionary-based text mining approach.

5 Preparing Data in R

The first step is to import and clean the data. The function `clean_names()` from the `janitor` package standardises variable names. The `select()` function keeps only the variables that are relevant for the analysis.

```
url <- paste0(
  "https://raw.githubusercontent.com/",
  "ruchitgandhi/Twitter-Airline-Sentiment-Analysis/",
  "master/Tweets.csv"
)

tweets <- readr::read_csv(url, show_col_types = FALSE) |>
  janitor::clean_names() |>
  select(
    airline_sentiment,
    negativereason,
    airline,
    retweet_count,
    text,
    tweet_created
  )
```

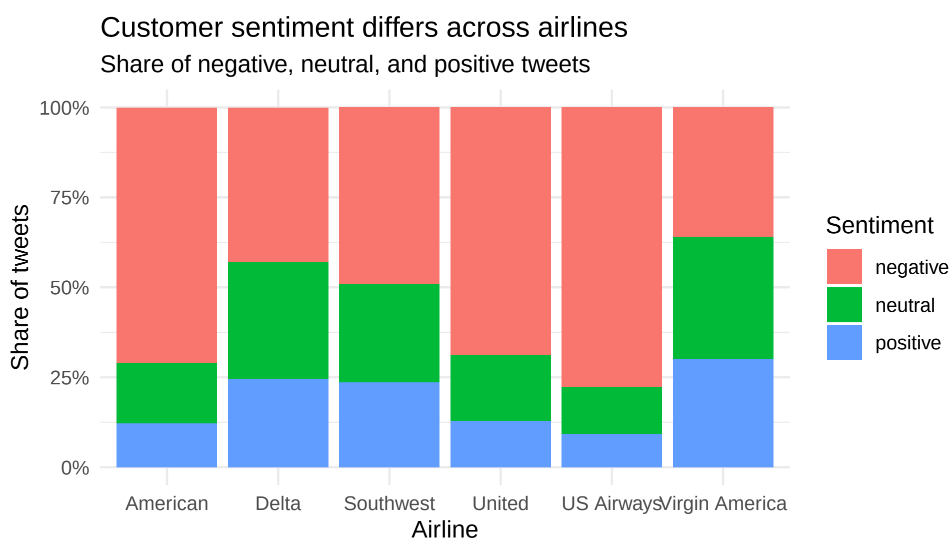
This step follows an important principle in reproducible data analysis: the workflow should be written in code so that the analysis can be repeated or adjusted later (Huber, 2025).

6 Descriptive Analysis of Sentiment

Before applying text mining, it is useful to examine the provided sentiment labels. The following figure shows how negative, neutral, and positive tweets are distributed across airlines.

```
sentiment_by_airline <- tweets |>
  count(airline, airline_sentiment) |>
  group_by(airline) |>
  mutate(share = n / sum(n)) |>
  ungroup()

ggplot(sentiment_by_airline, aes(x = airline, y = share, fill = airline_sentiment)) +
  geom_col(position = "fill") +
```

```r
  scale_y_continuous(labels = scales::percent) +
  labs(
    title = "Customer sentiment differs across airlines",
    subtitle = "Share of negative, neutral, and positive tweets",
    x = "Airline",
    y = "Share of tweets",
    fill = "Sentiment"
  ) +
  theme_minimal()
```



The figure is useful for decision-making because it allows managers to compare airlines quickly. However, a sentiment label alone is not enough. A manager also needs to understand why customers are dissatisfied.

7 Negative Sentiment and Business Problems

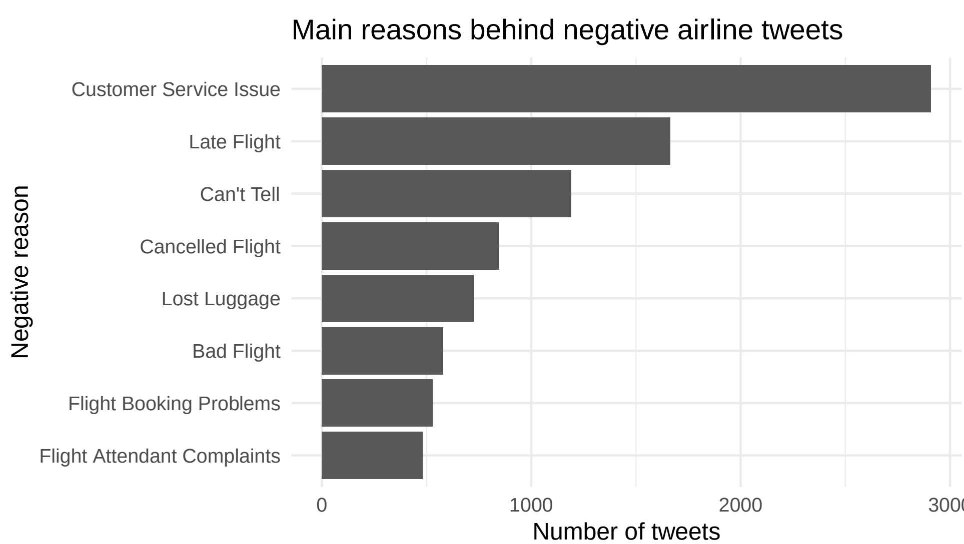
After the general sentiment overview, the analysis focuses on negative tweets. The dataset contains a variable that explains the reason behind many negative tweets. This is important because every negative reason can point to a different business problem.

```r
negative_reasons <- tweets |>
  filter(airline_sentiment == "negative", !is.na(negativereason)) |>
  count(negativereason, sort = TRUE)

negative_reasons |>
  slice_max(n, n = 8) |>
  knitr::kable()
```

negativereason	n
Customer Service Issue	2910
Late Flight	1665
Can't Tell	1190
Cancelled Flight	847
Lost Luggage	724
Bad Flight	580
Flight Booking Problems	529
Flight Attendant Complaints	481

The same result can also be visualised as a bar chart. This makes it easier to identify which business problems should receive attention first.

```r
negative_reasons |>
  slice_max(n, n = 8) |>
  ggplot(aes(x = reorder(negativereason, n), y = n)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Main reasons behind negative airline tweets",
    x = "Negative reason",
    y = "Number of tweets"
  ) +
  theme_minimal()
```



The results connect sentiment analysis to operational decision-making. Negative comments are not random; they often refer to service issues, delayed flights, cancelled flights, baggage problems, or booking problems. Each category suggests a different managerial response.

8 Text Mining and Tokenization

A central step in text mining is tokenization. Tokenization means splitting text into smaller units. In this example, each tweet is split into individual words. Stop words such as common function words are removed because they usually do not provide much meaning for sentiment analysis.

```
tweets_words <- tweets |>
  mutate(tweet_id = row_number()) |>
  select(tweet_id, airline, airline_sentiment, text) |>
  tidytext::unnest_tokens(word, text) |>
  anti_join(tidytext::stop_words, by = "word") |>
  filter(
    stringr::str_detect(word, "[a-z]"),
    !stringr::str_detect(word, "^http"),
    !stringr::str_detect(word, "^t.co")
  )

tweets_words |>
  count(word, sort = TRUE) |>
  head(10)
```

word	n
united	4154
flight	3926
usairways	3051
americanair	2962
southwestair	2457
jetblue	2366
cancelled	1064
service	965
time	791
customer	752

The tidy approach is useful because the result is a table in which each row is one word from one tweet. This format can be filtered, counted, joined with dictionaries, and visualised using familiar `dplyr` and `ggplot2` functions.

9 Lexicon-Based Sentiment Analysis

A lexicon-based method uses a dictionary that assigns sentiment labels to words. In this example, the Bing lexicon is used through the `get_sentiments("bing")` function. Words from the tweets are matched with positive and negative words from the dictionary.

```
word_sentiments <- tweets_words |>
  inner_join(
    tidytext::get_sentiments("bing"),
    by = "word"
```

```
  )

word_sentiments |>
  count(sentiment, sort = TRUE)
```

sentiment	n
negative	7841
positive	3965

This method is transparent and easy to explain. However, it has limitations. It may miss sarcasm, context, negation, and domain-specific meanings.

The following code shows the most frequently matched positive and negative sentiment words.

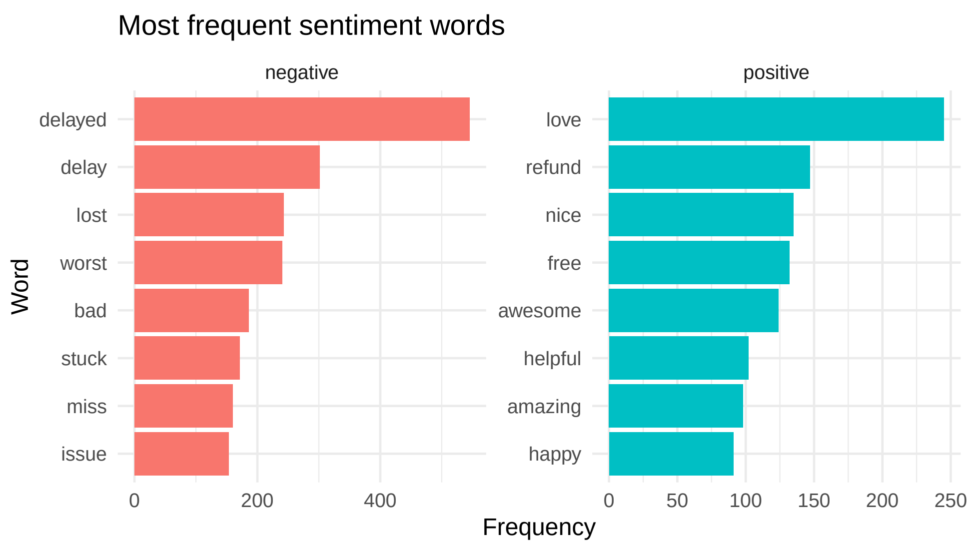
```
top_sentiment_words <- word_sentiments |>
  count(sentiment, word, sort = TRUE) |>
  group_by(sentiment) |>
  slice_max(n, n = 8) |>
  ungroup()

top_sentiment_words |>
  arrange(sentiment, desc(n))
```

sentiment	word	n
negative	delayed	545
negative	delay	301
negative	lost	243
negative	worst	240
negative	bad	186
negative	stuck	171
negative	miss	160
negative	issue	153
positive	love	245
positive	refund	147
positive	nice	135
positive	free	132
positive	awesome	124
positive	helpful	102
positive	amazing	98
positive	happy	91

The presentation also uses the following figure to show frequent positive and negative words visually.

```
top_sentiment_words |>
  mutate(word = reorder(word, n)) |>
```

```
  ggplot(aes(x = word, y = n, fill = sentiment)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ sentiment, scales = "free") +
  coord_flip() +
  labs(
    title = "Most frequent sentiment words",
    x = "Word",
    y = "Frequency"
  ) +
  theme_minimal()
```



The following code calculates a lexicon-based sentiment score for each airline.

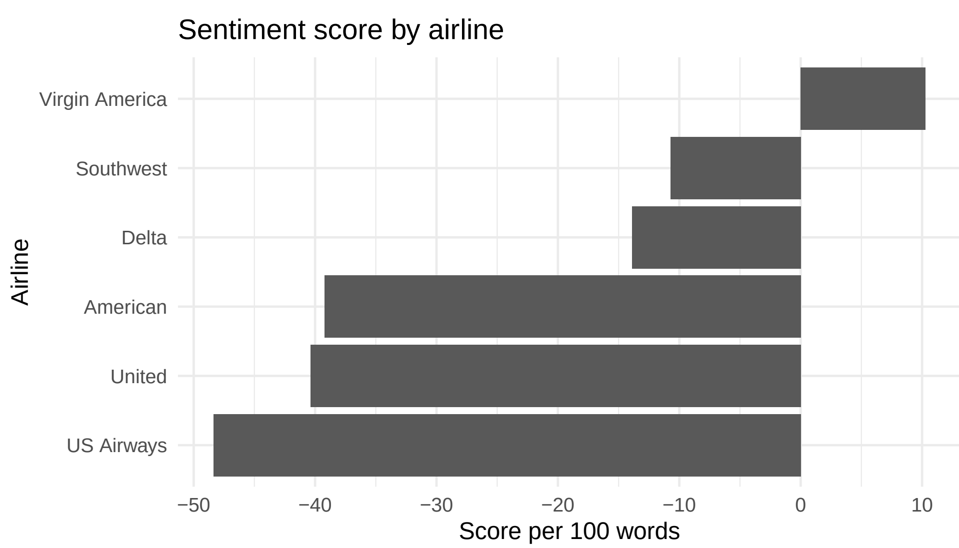
```
lexicon_airline_sentiment <- word_sentiments |>
  count(airline, sentiment) |>
  pivot_wider(
    names_from = sentiment,
    values_from = n,
    values_fill = 0
  ) |>
  mutate(
    lexicon_score = positive - negative,
    total_sentiment_words = positive + negative,
    lexicon_score_per_100_words = 100 * lexicon_score / total_sentiment_words
  ) |>
  arrange(lexicon_score_per_100_words)
```

airline	negative	positive	lexicon_score	total_sentiment_words	lexicon_score_per_100_words
US Airways	1925	670	-1255	2595	-48.36224

airline	negative	positive	lexicon_score	total_sentiment_words	lexicon_score_per_100_words
United	2432	1033	-1399	3465	-40.37518
American	1546	675	-871	2221	-39.21657
Delta	836	632	-204	1468	-13.89646
Southwest	945	762	-183	1707	-10.72056
Virgin America	157	193	36	350	10.28571

The score is calculated as positive words minus negative words, standardized per 100 sentiment words.

```
ggplot(
  lexicon_airline_sentiment,
  aes(
    x = reorder(airline, lexicon_score_per_100_words),
    y = lexicon_score_per_100_words
  )
) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Sentiment score by airline",
    x = "Airline",
    y = "Score per 100 words"
  ) +
  theme_minimal()
```



10 From Analysis to Managerial Action

The final step is to translate analytical results into decisions. A manager should not only ask whether sentiment is positive or negative. A better question is: What action should this result trigger?

```r
decision_actions <- negative_reasons |>
  mutate(
    managerial_action = case_when(
      negativereason == "Customer Service Issue" ~
        "Improve support capacity.",
      negativereason == "Late Flight" ~
        "Improve delay communication.",
      negativereason == "Cancelled Flight" ~
        "Strengthen rebooking process.",
      negativereason == "Lost Luggage" ~
        "Improve baggage tracking.",
      negativereason == "Bad Flight" ~
        "Improve in-flight experience.",
      TRUE ~ "Investigate further."
    )
  )

decision_actions |>
  select(negativereason, n, managerial_action) |>
  head(8)
```

negativereason	n	managerial_action
Customer Service Issue	2910	Improve support capacity.
Late Flight	1665	Improve delay communication.
Can't Tell	1190	Investigate further.
Cancelled Flight	847	Strengthen rebooking process.
Lost Luggage	724	Improve baggage tracking.
Bad Flight	580	Improve in-flight experience.
Flight Booking Problems	529	Investigate further.
Flight Attendant Complaints	481	Investigate further.

This table shows how sentiment analysis can move from description to decision support. Customer service problems suggest a different response than lost luggage or delayed flights. In this sense, sentiment analysis can help managers prioritize resources.

11 Exercise for Students

Run the following R code and answer the question below.

```r
library(tidyverse)
library(tidytext)

customer_comments <- tibble(
  text = c(
    "The flight was delayed and the service was terrible",
    "The staff was helpful and the flight was comfortable",
```

```
    "My luggage was lost and nobody helped me"
  )
)

customer_comments |>
  unnest_tokens(word, text) |>
  anti_join(stop_words, by = "word") |>
  inner_join(get_sentiments("bing"), by = "word") |>
  count(sentiment)
```

Question: Which managerial action would be most suitable based on the negative words in the comments?

A) Reduce the number of customer support channels

B) Improve delay communication and luggage recovery processes

C) Ignore negative comments because they are unstructured data

D) Only focus on positive words

Correct answer: B) Improve delay communication and luggage recovery processes.

12 Limitations

Sentiment analysis is useful, but it should not be interpreted as a perfect measurement of customer opinion. Lexicon-based approaches are simple and explainable, but they may fail when language is ironic, ambiguous, or context-dependent. Short social media posts are especially challenging because they often contain abbreviations, slang, or missing context.

For managerial decision-making, sentiment analysis should therefore be used as a support tool rather than as a replacement for human judgment. Managers should combine sentiment results with operational indicators, customer service data, and qualitative interpretation.

13 Conclusion

Sentiment analysis helps transform unstructured customer feedback into information that can support decision-making. In the airline example, R is used to import tweets, clean data, summarize sentiment labels, tokenize text, apply a sentiment lexicon, and connect complaint categories to managerial actions.

The main lesson is that the value of sentiment analysis is not the sentiment label itself. The value is the decision that follows from it. When text analysis identifies repeated customer pain points, managers can prioritize concrete improvements in service, communication, booking processes, or baggage handling.

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Checklist:

- ☒ The handout contains 3-5 pages of text.
- ☒ The submission contains the Quarto file of the handout.
- ☒ The submission contains the Quarto file of the presentation.
- ☒ The submission contains the HTML file of the handout.
- ☒ The submission contains the HTML file of the presentation.
- ☒ The submission contains the PDF file of the handout.
- ☒ The submission contains the PDF file of the presentation.
- ☒ The title page of the presentation and the handout contain personal details: name, email, and matriculation number.
- ☒ The handout contains a bibliography, created using BibTeX with an APA citation style.
- ☒ Either the handout or the presentation contains R code that proves the expertise in coding.
- ☒ The filled out Affidavit.
- ☒ The link to the presentation and the handout published on GitHub.

Mustafa Asım Ruhi, 2 June 2026, Cologne

14 References

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